Lecture 6

Lemmatization

Lemmatization means finding the original base word (dictionary form) of a word, even if it looks different.

Example:

- am, are, is → **be**
- car, cars, car's → **car**

Real life example:

When searching online, if you search **"running"**, the system understands it as **"run"** so it can show better results.

It helps computers know that different forms of a word have the same meaning.

Stemming

Stemming is a simpler method than lemmatization.
It just removes the ending part of a word without checking the correct dictionary form.

Example:

- playing → **play**
- studies → **studi**

("studi" is not a real word, so stemming is less accurate)

Real life example:

Search engines use stemming to quickly match words like **connected**, **connecting**, and **connection**.

Sentence Segmentation

Sentence segmentation means dividing a paragraph into separate sentences.

Example:

Paragraph:
Shelby is a student. She is sick today. She will not go to school.

After segmentation:

- Shelby is a student
- She is sick today
- She will not go to school

Real life example:

Mobile phones and chat apps use sentence segmentation to understand where one sentence ends and the next starts.

Note:

- **?** and **!** are easy sentence endings
- **.** is sometimes confusing because of:
    - Dr. Ahmed
    - Inc.
    - 4.3
    - 0.02%

Lecture 8

1. Word-based Tokenization

It means splitting text into words using spaces and punctuation.

Example:

"I love cats" → **I | love | cats**

Real life example:

Google search splits your sentence into separate words to understand your search.

Advantages:

- Very easy to use
- Easy to understand

Disadvantages:

- "Let" and "Let's" become different words
- Wrong spelling becomes a new token
- Chinese language has no spaces
- Vocabulary becomes very large

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2. Character-based Tokenization

It means splitting text into single characters.

Example:

"Cat" → **C | a | t**

Real life example:

Used in password checking or spelling correction systems.

Advantages:

- Almost no unknown words
- Good for languages where characters have meaning
- Easy to apply

Disadvantages:

- One character usually has no full meaning
- Hard to understand full word meaning
- More input for computer to process

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3. Subword Tokenization

It splits rare words into smaller meaningful parts but keeps common words complete.

Example:

"Tokenization" → **Token | ##ization**

Real life example:

Used in AI tools like chatbots and translators.

Advantages:

- Solves unknown word problem
- Smaller vocabulary size

Disadvantages:

- More complex method

Examples:

BPE and **WordPiece**

Byte Pair Encoding (BPE)

BPE is a subword tokenization method.

It starts from single characters and keeps joining the most common pair.

Training Example

Words:

- low
- lowest
- newer
- wider
- new

First Vocabulary:

_, d, e, i, l, n, o, r, s, t, w

(_ means end of word)

Most Frequent Pair:

e + r = er

Because “er” appears 9 times.

So new vocabulary becomes:

_, d, e, i, l, n, o, r, s, t, w, er

This is the final vocabulary after this merge.

Using BPE

Example 1:

Input: **newer**

Tokens:
newer

because full merge exists

Example 2:

Input: **lower**

Tokens:
low | er

because “low” and “er” are learned tokens

Stopwords Removal

Stopwords are very common words like:

the, of, and, to

These words usually do not add important meaning.

Example:

“I am going to the school”

After removing stopwords:
going school

Real life example:

Search engines remove stopwords to save space and improve speed.

Problem:

Sometimes stopwords are important

Example:

- “to be or not to be”
- “flights to Portland”

So we must be careful.

Token / Term Normalization

It means making different forms of a word look the same.

Example:

- U.S.A → USA
- color → colour

- anti-discriminatory → antidiscriminatory

Real life example:

Google understands “color” and “colour” as same meaning.

Stemming

Stemming means removing word endings to get the root form.

Example:

- fish
- fishes
- fishing

→ **fish**

Real life example:

If you search “fishing”, the system can also show results for “fish”.

Types:

- Aggressive → fish and fishing same
- Conservative → only remove simple plural “s”
- No stemming → keep all words separate

Problem:

Sometimes wrong results happen

Example:
Centuries → Centurie

(not a correct word)

Lecture 9

Stemming

Stemming means reducing words to a common root (stem).

Example:

- fish
- fishes
- fishing

→ **fish**

It helps group similar words together.

Real life example:

If you search **“fishing”**, the search engine may also show results for **“fish”**.

Types of Stemming

1. Aggressive Stemming

It removes more letters.

Example:
fish and fishing → **fish**

2. Conservative Stemming

It removes only small endings like plural **s**

Example:
cats → **cat**

3. No Stemming

All words stay separate

Example:
fish, fishes, fishing all remain different

Problems in Stemming

Over-stemming

Too much part of the word is removed.

Example:

- centennial
- century
- center

→ all become **cent** (wrong)

Under-stemming

Too little part is removed.

Example:

- acquire
- acquired → acquir
- acquisition → acquis

These should be grouped but are different.

Porter Stemmer

It is the most common English stemmer made by **Martin Porter**.

It removes endings like:

- ing
- sses
- eed

Example:
agreed → agree

Advantage:

- Better results than many other stemmers
- Less error rate

Disadvantage:

- Sometimes gives stems that are not real words

Krovetz Stemmer

Made by **Robert Krovetz**

It first checks the dictionary.

If word exists → keep it
If not → remove suffix and check again

Advantage:

- Produces real words, not broken stems

Disadvantage:

- Can miss some correct matches

FP, FN, TP, TN

These are used to check system accuracy.

True Positive (TP)

System says correct, and it is correct.

Example:

Spam email is detected as spam.

True Negative (TN)

System says not correct, and it is really not correct.

Example:

Normal email is detected as normal.

False Positive (FP)

System says correct, but it is wrong.

Example:

Policy → Police (wrong match)

A normal email marked as spam.

False Negative (FN)

System says wrong, but it should be correct.

Example:

noise and noisy are not matched

Spam email marked as normal.

Lemmatization

Lemmatization means finding the correct dictionary base form of a word.

Example:

- am, are, is → **be**
- car, cars, car's → **car**

Sentence Example:

"The boy's cars are different colors"

→ **the boy car be different color**

Real life example:

When searching **"running"**, the system understands it as **"run"** for better results.

Difference from Stemming:

- Stemming cuts letters only
- Lemmatization finds the correct real word

Phrases

A phrase is a group of words used together.

Example:

- red apple
- black sea
- University of Southern Maine

Phrases are more clear than single words.

Example:
apple is general
red apple is more specific

Real life example:

Google search gives better results for **"red shoes"** than only **"shoes"**

N-gram

N-gram means a group of words together.

Types:

Unigram

1 word

Example:
apple

Bigram

2 words

Example:
red apple

Trigram

3 words

Example:
I love cats

The more a phrase repeats, the more meaningful it is likely to be.

Lecture 10

Text Classification (Text Categorization)

Text classification means giving a label or category to a full text or document.

Example:

Email → Spam or Not Spam
Movie review → Positive or Negative

Common Text Classification Tasks

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1. Sentiment Analysis

It finds whether the text feeling is **positive** or **negative**.

Example:

“This movie is amazing!” → **Positive**
“This pizza is awful” → **Negative**

Real life example:

Online shopping websites check customer reviews to know if people like a product.

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2. Spam Detection

It checks whether an email is **spam** or **normal**.

Example:

“You won 1 million dollars!” → **Spam**

Real life example:

Gmail automatically puts fake promotion emails into spam folder.

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3. Authorship Identification

It finds who wrote the text.

Example:

Finding whether a poem was written by a specific writer.

Real life example:

Used in crime investigation for unknown messages.

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4. Age / Gender Identification

It guesses writer details like age or gender.

Example:

Text style may show if writer is young or old.

Real life example:

Used in marketing to understand customers.

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5. Language Identification

It finds which language is used.

Example:

“Bonjour” → French

Real life example:

Google Translate first detects the language automatically.

Simple Sentiment Analysis

This is usually **Binary Classification**

Only two classes:

- Positive (+)
- Negative (-)

Example:

“Awesome food, I love this place!” → **Positive**

“Awful pizza and overpriced” → **Negative**

Why Sentiment Analysis?

It helps in many areas:

Movies

Is review positive or negative?

Products

Do people like the new iPhone?

Politics

What do people think about a candidate?

Prediction

Can public opinion help predict elections or market trends?

Text Classification Formula

Input:

- A document (text)
- A fixed set of classes

Example:
Classes = {Spam, Not Spam}

Output:

- One predicted class

Example:
Email → Spam

Classification Methods

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1. Hand-coded Rules

Humans write rules manually.

Example:

If email contains:

- “dollars”
- “you have been selected”

→ classify as **Spam**

Advantage:

- Can be accurate

Disadvantage:

- Expensive to make
- Hard to maintain
- Rules can break easily

Real life example:

Old email filters used rule-based spam detection.

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2. Supervised Machine Learning

The computer learns from already labeled examples.

Example:

Many movie reviews already marked as positive or negative

The model learns from them and predicts for new reviews.

Real life example:

Netflix review systems use this idea.

Some Classifiers

- Naïve Bayes
- Logistic Regression

- Support Vector Machine (SVM)
- k-Nearest Neighbors (KNN)

Naïve Bayes Classifier

It is based on **Bayes Rule**.

It assumes features are simple and independent.

It uses probability to decide the class.

Bag of Words

A document is treated like a bag of words.

Word order is ignored.
Only word frequency matters.

Example:

“I love pizza”
“Pizza love I”

Both are treated almost the same.

Real life example:

Spam detection often uses bag of words.

Problem with MLE

If a word never appeared in training data, probability becomes **zero**.

Example:

Word = fantastic

If it never appeared in positive reviews, system may fail.

Zero probability creates problems.

Solution: Add-One (Laplace) Smoothing

We add 1 to every word count.

This avoids zero probability.

Example:

Even if “fantastic” never appeared, it still gets a small probability.

Unknown Words Problem

Sometimes test data has a word not in vocabulary.

Example:

Training data has no word:
“superb”

Test review contains:
“This phone is superb”

Simple solution:

Remove that unknown word from test document.